

Year 8 Digital Technologies

Unit 2 — Wearable Technology

St Brendan's College (Yeppoon), 2026



Unit overview		Cohort and/or class considerations
Start week: Term 1 Week 1 Finish week: Term 2 Week 10 In this unit, students use Micro:bit devices to explore data, communication and physical computing in a practical and age-appropriate context. Students learn how Microbits collect, transmit and represent data, with a focus on radio communication, binary representation and basic data security concepts. Through guided investigations, students explore how data can be gathered from sensors, transmitted between devices and analysed to identify patterns and trends. In the second part of the unit, students apply these skills to design and create wearable technologies, combining computational thinking, data analysis and ethical considerations when collecting data about people.		
Adjustments		
Student initials/identifier		
Year level/band against which the student is to be assessed	Band 7–8	
Category of need		
Adjustment/s and/or key information		
Review of adjustments		

Assessment 2	
Year 8	
Title	Data Exam
Description	In this assessment, students complete a written exam that assesses their understanding of how Microbits communicate and represent data. Students explain the differences between broadcast and direct radio communication, and demonstrate their understanding of how text, numbers and messages are represented using integers and binary. The exam also assesses students' ability to model and reason about structured data, design and trace basic algorithms, and predict program outputs related to Microbit communication and simple encryption. This assessment provides evidence of students' conceptual understanding of data transmission, representation and algorithmic thinking.
Technique	Examination
Mode	Written
Conditions	• up to 50 minutes plus 10 minutes planning time • under supervised conditions • short response up to 75 words per item, up to 400 words for the task • extended response 100–200 words per item, up to 400 words for the task
Timing	Week 7
Achievement standard	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.
Moderation	Cross marked within the department

Content descriptions	
Knowledge and understanding	Processes and production skills
Digital systems investigate how data is transmitted and secured in wired and wireless networks including the internet (AC9TDI8K02)	Acquiring, managing and analysing data model and query the attributes of objects and events using structured data (AC9TDI8P03)

Content descriptions	
Knowledge and understanding	Processes and production skills
Data representation investigate how digital systems represent text, image and audio data using integers (AC9TDI8K03)	Generating and designing design algorithms involving nested control structures and represent them using flowcharts and pseudocode (AC9TDI8P05)
explain how and why digital systems represent integers in binary (AC9TDI8K04)	trace algorithms to predict output for a given input and to identify errors (AC9TDI8P06)

Assessment 3	
Year 8	
Title	Wearable Tech
Description	In this assessment, students design and develop a wearable Microbit-based solution to collect and analyse data related to a given investigation task. Students plan how data will be gathered using appropriate sensors, design algorithms to collect and store the data, and use digital tools to analyse and visualise results to draw conclusions. They select suitable hardware, design the user experience, manage the project process and evaluate their solution against design criteria and user needs. Students also demonstrate responsible and ethical data collection practices, reflecting on the purpose and impact of their digital solution.
Technique	Investigation
Mode	Multimodal
Conditions	2–4 A3 pages or equivalent digital media pages that may include annotated graphical representations
Timing	Week 9
Achievement standard	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.
Moderation	Monitoring: Week 6 Conferencing: Week 8

Content descriptions	
Knowledge and understanding	Processes and production skills
Digital systems explain how hardware specifications affect performance and select appropriate hardware for particular tasks and workloads (AC9TDI8K01)	Acquiring, managing and analysing data acquire, store and validate data from a range of sources using software, including spreadsheets and databases (AC9TDI8P01)
investigate how data is transmitted and secured in wired and wireless networks including the internet (AC9TDI8K02)	analyse and visualise data using a range of software, including spreadsheets and databases, to draw conclusions and make predictions by identifying trends (AC9TDI8P02)
	model and query the attributes of objects and events using structured data (AC9TDI8P03)
	Investigating and defining define and decompose real-world problems with design criteria and by creating user stories (AC9TDI8P04)
	Generating and designing design algorithms involving nested control structures and represent them using flowcharts and pseudocode (AC9TDI8P05)
	trace algorithms to predict output for a given input and to identify errors (AC9TDI8P06)
	design the user experience of a digital system (AC9TDI8P07)
	generate, modify, communicate and evaluate alternative designs (AC9TDI8P08)
	Producing and implementing implement, modify and debug programs involving control structures and functions in a general-purpose programming language (AC9TDI8P09)
	Evaluating evaluate existing and student solutions against the design criteria, user stories and possible future impact (AC9TDI8P10)

Content descriptions	
Knowledge and understanding	Processes and production skills
	Collaborating and managing select and use a range of digital tools efficiently, including unfamiliar features, to create, locate and communicate content, consistently applying common conventions (AC9TDI8P11)
	select and use a range of digital tools efficiently and responsibly to share content online, and plan and manage individual and collaborative agile projects (AC9TDI8P12)
	Privacy and security investigate and manage the digital footprint existing systems and student solutions collect and assess if the data is essential to their purpose (AC9TDI8P14)
General capabilities	
Critical and creative thinking	
• Inquiring - Develop questions - Identify, process and evaluate information • Generating - Create possibilities - Consider alternatives - Put ideas into action • Analysing - Interpret concepts and problems - Draw conclusions and provide reasons - Evaluate actions and outcomes	

Digital literacy

- Practising digital safety and wellbeing
 - Manage digital privacy and identity
- Investigating
 - Locate information
 - Acquire and collate data
 - Interpret data

Ethical understanding

Literacy

- Reading and viewing
 - Fluency
- Writing
 - Creating texts
 - Handwriting and keyboarding

Numeracy

- Number sense and algebra
 - Number and place value
- Measurement and geometry
- Statistics and probability
 - Understanding chance
 - Interpreting and representing data

Teaching and learning sequence

Week	Planning details	Theory	Prac	Assessment
1	Key teaching and learning experiences	Introduction to data in real-world contexts. What is data, where it comes from, and why we collect it. Overview of Micro:bit as	Students set up Micro:bit and run simple programs to collect live data (e.g. temperature, light, movement). Begin	Introduce Assessment 2 (Data Exam – Week 7). Outline key concepts (data, representation, communication). Diagnostic observation of student

Teaching and learning sequence				
		a data collection device (sensors, inputs, outputs).	streaming live data to Excel and observe how data appears in tabular form.	confidence with devices and Excel interface.
	Adjustments			
	Resources	Micro:bit online tutorials (Getting Started, Sensors) ORB Education Excel documents (intro spreadsheets) Micro:bit + USB devices		
2	Key teaching and learning experiences	Types of data (numerical, categorical, Boolean). Structured data and why spreadsheets use rows/columns. Link between collected sensor data and structured datasets.	Students collect a controlled dataset using Micro:bit (e.g. temperature over time). Export and/or stream into Excel. Format data tables correctly (headings, units, consistent structure).	Checkpoint: students submit a clean dataset (correct structure and headings). Teacher feedback on data quality and organisation.
	Adjustments			
	Resources	Micro:bit tutorials (data logging / streaming) ORB Education Excel documents (data entry and formatting)		
3	Key teaching and learning experiences	Data quality and validation. Accuracy, consistency, and errors in data collection. Why validation is important in digital systems.	Use Excel data validation tools (drop-down lists, number limits) on collected Micro:bit datasets. Students refine their dataset to remove or prevent errors.	Formative task: students submit a validated dataset with applied rules. Feedback focused on accuracy and control.
	Adjustments			
	Resources	ORB Education Excel documents (data validation) Micro:bit (repeat data collection for testing validation rules)		

Teaching and learning sequence				
4	Key teaching and learning experiences	Introduction to data analysis. Identifying patterns and trends in datasets. Choosing appropriate ways to represent data visually.	Students use their Micro:bit dataset to create graphs in Excel (line graphs for time data, column graphs for comparisons). Interpret patterns from their own collected data.	Mini task: create one graph and write a short interpretation of the data. Ongoing monitoring toward exam concepts.
	Adjustments			
	Resources	ORB Education Excel documents (charts and graphs) Student-generated Micro:bit datasets		
5	Key teaching and learning experiences	Data representation in digital systems. Introduction to integers and binary (link to how Micro:bit sends/stores data).	Simple binary activities (encoding numbers/messages). Link back to Micro:bit by discussing how sensor values are represented digitally. Continue refining Excel datasets and graphs.	Exam preparation begins – short practice questions on data representation and structured data.
	Adjustments			
	Resources	Micro:bit tutorials (basic communication concepts) Teacher-created binary activities ORB Education Excel documents (review tasks)		
6	Key teaching and learning experiences	Data communication concepts. How Micro:bits transmit data (radio communication – broadcast vs direct). Introduction to reliability and basic security ideas.	Students program Micro:bits to send data between devices. Optionally stream received data into Excel for comparison/analysis.	Checkpoint: students explain (written or verbal) how their Micro:bits are collecting and transmitting data. Final formative check before exam.
	Adjustments			
	Resources	Micro:bit tutorials (radio communication) ORB Education Excel documents (data comparison tasks)		

Teaching and learning sequence				
7	Key teaching and learning experiences	Revision of key concepts for Assessment 2, including structured data, data validation, data representation (integers and binary), and data communication (broadcast vs direct). Explicit teaching of exam strategies, including interpreting questions, structuring short and extended responses, and using correct terminology.	Students complete guided revision activities, including practice exam questions and collaborative review tasks. Use Micro:bit examples and previously created Excel datasets to reinforce concepts.	Assessment 2 – Data Exam conducted under exam conditions.
	Adjustments			
	Resources	Practice exam questions (teacher-created) Previous student datasets (Excel) Micro:bit devices for demonstration		
8	Key teaching and learning experiences	Introduction to algorithms and computational thinking. Understanding sequence, selection and iteration in simple programs.	Students create simple Micro:bit programs using control structures (loops, if statements). Test and modify programs.	Formative monitoring of student ability to design and implement simple algorithms.
	Adjustments			
	Resources	Micro:bit online tutorials (control structures, loops) Teacher exemplars		
9	Key teaching and learning experiences	Tracing and evaluating algorithms. Predicting outputs and identifying errors.	Students trace given Micro:bit programs and debug faulty code. Modify programs to achieve desired outputs.	Short applied task: students explain how a given program works and identify errors.
	Adjustments			
	Resources	Micro:bit tutorials (debugging)		

Teaching and learning sequence				
		Teacher-created tracing worksheets		
10	Key teaching and learning experiences	Data transmission in networks. How Micro:bits communicate using radio (broadcast vs direct communication).	Students program Micro:bits to send and receive messages between devices. Experiment with broadcast and targeted communication.	Observation and questioning to assess understanding of communication types.
	Adjustments			
	Resources	Micro:bit tutorials (radio communication) Class set of Micro:bits		
11	Key teaching and learning experiences	Data security and reliability. Introduction to simple encryption concepts and why secure communication matters.	Students implement simple encoded messages between Micro:bits. Test reliability of communication.	Formative task: students demonstrate and explain a secure or encoded communication example.
	Adjustments			
	Resources	Micro:bit tutorials (radio + variables) Teacher-created encryption activities		
12	Key teaching and learning experiences	Introduction to wearable technology. Exploring real-world examples and defining user needs and purposes.	Students investigate examples of wearable tech and brainstorm possible project ideas linked to data collection.	Assessment 3 introduced. Task sheet unpacked, criteria explained, examples analysed.
	Adjustments			
	Resources	Teacher slides/examples of wearable tech Micro:bit project gallery Assessment task sheet		
13	Key teaching and learning experiences	Defining problems and user stories. Identifying data needs and ethical considerations (privacy, relevance of data).	Students develop a project idea and write a simple user story. Begin planning how data will be collected using Micro:bit sensors.	Submission of initial project proposal for feedback.

Teaching and learning sequence				
	Adjustments			
	Resources	Planning templates Micro:bit sensor guides		
14	Key teaching and learning experiences	Designing algorithms and systems. Representing ideas using flowcharts and pseudocode.	Students design the logic for their wearable solution (flowcharts/ pseudocode) and begin initial coding.	
	Adjustments			
	Resources	Flowchart templates Micro:bit tutorials (advanced inputs/ outputs)		
15	Key teaching and learning experiences	Prototyping and iterative design. Testing and refining solutions.	Students begin building their wearable prototype using Micro:bit and materials.	Progress check against design criteria.
	Adjustments			
	Resources	Micro:bits, craft/electronics materials Teacher exemplars		
16	Key teaching and learning experiences	Data collection and analysis in projects. Ensuring accuracy and relevance of collected data.	Students test their wearable device and collect sample data. Analyse results using Excel.	Checkpoint: evidence of data collection and initial analysis.
	Adjustments			
	Resources	ORB Education Excel documents (analysis tasks) Micro:bit devices		

Teaching and learning sequence				
17	Key teaching and learning experiences	Debugging and improving solutions. Evaluating effectiveness against criteria.	Students refine code and physical design of their wearable solution.	Teacher conferencing on progress and improvements.
	Adjustments			
	Resources	Micro:bit tutorials (debugging) Teacher feedback notes		
18	Key teaching and learning experiences	User experience and design impact. Considering usability and ethical implications.	Students finalise wearable design and ensure functionality meets user needs.	Draft submission or prototype demonstration.
	Adjustments			
	Resources	Evaluation checklists Peer feedback tools		
19	Key teaching and learning experiences	Communicating digital solutions. Structuring explanations and justifications.	Students prepare final presentation (multimodal or demonstration).	Final submission of Assessment 3.
	Adjustments			
	Resources	Presentation templates Rubric/criteria sheet		
20	Key teaching and learning experiences	Reflection on learning across the unit. Connections between data, systems, and design.	Students present projects and participate in peer evaluation.	Reflection task and final evaluation of project.
	Adjustments			
	Resources	Reflection prompts		

Teaching and learning sequence				
		Peer assessment sheets		
Evaluation				

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